

Name: TAYLOR, VERSIONStudent ID: RUBRIC

ECON 0150 | MiniExam 3 | Spring 2026

This MiniExam will take 16 minutes with a quick break to follow. MiniExams are designed to both test your knowledge and challenge you to apply familiar concepts in new environments. Treat it as if you're trying to show me that you understand the material. Answer clearly, completely, and concisely.

Academic Conduct Code

10 pts The following academic conduct code is designed to protect the integrity of your work. Print your name/initials beside the three academic honesty agreements. I pledge to my fellow students, the university, and the instructor, that:

- ___ I will complete this MiniExam solely using my own work.
___ I will not use any digital resources unless explicitly allowed by the instructor.
___ I will not communicate directly or indirectly with others during the MiniExam.

25 pts Q1. A class's average sleep time follows an *exponential* distribution with mean (μ) of 7.6 hours and standard deviation (σ) of 1 hour.

a) If you sample one student ($n = 1$), what is the sampling distribution of the sample means?

Shape: Exponential

Mean: 7.6

Standard error: 1

Note: Many students answered "Uniform" which only makes sense if they copy from the Demo. This tells me they haven't demonstrated learning the CLT. I think everyone gave me "Normal" for the other two.

b) If you sample 49 students ($n = 49$), what is the sampling distribution of the sample means?

Shape: Normal

Mean: 7.6

Standard error: $\frac{1}{\sqrt{49}} = \frac{1}{7}$

c) If you sample one student ($n = 144$), what is the sampling distribution of the sample means?

Shape: Normal

Mean: 7.6

Standard error: $\frac{1}{\sqrt{144}} = \frac{1}{12}$

Q2. You take 10,000 samples ($n = 64$) of classes, each with sleep times following an exponential distribution with mean (μ) of 7.6 hours and standard deviation (σ) of 1 hours (same as in Q1).

a) If you were to plot a histogram of the sample means student's sleep times, describe the distribution you would expect to see:

Shape: Normal

Mean: 7.6

Standard error: $\frac{1}{\sqrt{64}} = \frac{1}{8}$

↔ Grade as a whole

b) Your friend takes 10,000 samples of $n = 16$ instead of $n = 64$. Compared to your histogram from part (a), their histogram of sample means would be (select one):

☒ Wider and centered at 7.6

No partial credit.

☐ Narrower and centered at 7.6

☐ Wider and centered at a different value

☐ The same width but different shape

c) You draw one sample of 64 students and get a mean of 8.1 hours. Using your answer from part (a), explain in one sentence whether or not this result is surprising.

8.1 hours is 0.5 hours away from 7.6 hours, which is many $SE = \frac{1}{8}$ away from 7.6, so its very surprising.

Q3. You are interested in whether students sleep on average 8 hours per night. You take one sample of 100 students ($n = 100$) with an average sleep time ($\bar{x}$) of 7.5 hours and standard deviation (S) of 1 hour.

a) Describe the sampling distribution under the null hypothesis.

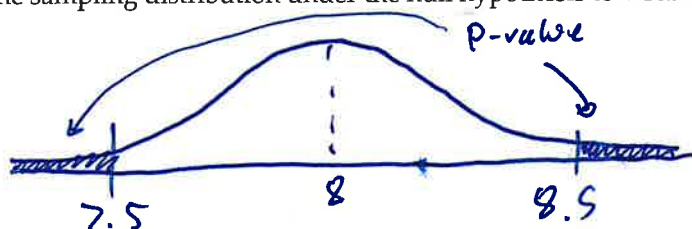
Shape: Normal

Mean: 8 ← Partial credit if 7.5

Standard error: $\frac{1}{10}$

b) Construct a confidence interval using 2 standard errors. Lower: 7.8 Upper: 8.2

c) Use a figure of the sampling distribution under the null hypothesis to visualize the p-value for your test.



Q4. A researcher conducts a test of whether students sleep on average 8 hours per night with a p-value of 0.12. Describe in one sentence what this p-value means.

This p-value tells us that the probability of seeing a sample mean at least as extreme as our sample is 0.12 if the null hypothesis were true.

Q1. 25 pts

10 pts SHAPE (a-c)

+10 Excellent if all correct.

+2 Progressing if "Uniform" in a)

+0 Blank

5 pts MEAN (a-c)

+5 Excellent if all correct.

+2 Progressing if 2 correct.

+1 Incomplete if less than 2 correct.

10 pts SE (a-c)

+10 Excellent if all correct.

+9 Satisfactory if small math error but good formula.

+6 Progressing if some meaningful math errors.

+1 Incomplete if not using $\sigma/\sqrt{n}$.

Q2. 25 pts

a) 9 pts +9 Excellent if all correct

+8 Satisfactory if small corrections only (probably won't use this one)

+6 Progressing if some meaningful corrections

+2 Incomplete if many corrections

8 pts b) +8 Correct

+0 Incorrect

18 pts c) +8 Excellent if mentions 1) distance between sample and mean, 2) SE, 3) surprising

+7 Satisfactory if small error or omission.

+2 Incomplete if multiple issues

+3 Progressing if meaningful error or omission

Q3. 25 pts

9pts a) Same as Q2.a except that a mean = 7.5 should be considered a small correction only ~~also~~

6pts b) ⁺⁶ Excellent if correct

+4 Progressing if (6,10) or (7.3,7.7) or (9.5,9.9)

+1 Incomplete if anything else

c) ⁺¹⁰ Excellent if includes all 5 correctly

10pts ⁺⁸ Satisfactory if includes 4 correctly

+5 Progressing if 3

+2 Incomplete if 2

Q4. 15 pts

+15 Excellent if all 5

+12 Satisfactory if 4

+5 Progressing if 3

+2 Incomplete if 2